MLB Championship Success:

Data Analysis on Determining Play-Off Metrics

Ranson Touch

Oregon State University

CS 332 – Intro to Applied Data Science

Computer Science Major

touchr@oregonstate.edu

Table of Contents

Introduction 1
Data Gathering 5
Direct Data Download(s) 5
Data Gathering Using an API 7
Data Cleaning 11
Los Angeles Dodgers Regular Season Batting Data 11
Los Angeles Dodgers Batting Post-Season Stats 16
Unsupervised Learning with Kmeans Clustering 22
Postseason Dodger Batter Stats 22
K = 2 25
K = 3 27
K = 4 29
Conclusions 31
Supervised Learning with Decision Trees 32
Conclusion 38
Conclusions 40
Reference 43

Introduction



Figure 1: The "GOAT" of baseball
Source: Fanatics

The MLB Postseason is often described as a different game from the 162-game regular season—pitching changes shorten, bullpen leverage spikes, and small edges on the margins (defense, baserunning, matchups) loom large. Regular-season dominance doesn't always translate into October success, which makes identifying the metrics that actually move series outcomes both interesting and useful. Fans are used to seeing "100-win teams" get bounced early while supposedly less impressive clubs catch fire at just the right time, which only adds to the mystery of what really works in October.

I chose this topic because I love baseball analytics and want to understand which measurable factors separate World Series contenders from early exits. As a fan, I've watched

teams with elite rotations ride pitching to a title and others win by combining on-base skills, power, and airtight defense. Those contrasts led me to ask: what quantifiable elements best predict postseason advancement in MLB? Is it mainly run prevention, overwhelming lineups, or some balance of the two that helps a team keep moving through each round?

This question matters to a lot of different people. Front offices and coaching staffs are constantly trying to build rosters that are not only good from April to September, but also built to survive the chaos of October. Players feel the difference too—one clutch swing or one defensive mistake can define how their entire season is remembered. And fans, of course, live and die with each playoff game, arguing about whether their team "has the right mix" to actually win it all. Having a clearer sense of which on-field strengths tend to show up in deeper runs can help everyone talk about postseason performance in a way that goes beyond vague ideas like "clutch" or "momentum."

To make this big question more concrete, this project looks at patterns in postseason performance over multiple years and asks what kinds of strengths tend to show up in teams that advance. Do successful playoff teams lean more on shutdown pitching staffs, or do they usually have lineups that combine getting on base with real power? Are tight, low-scoring wins more common for teams that go far, or do future champions tend to outslug their opponents? By framing the topic this way, the goal is to connect the drama of October baseball with specific, measurable on-field traits that fans and decision-makers already care about.

In the end, this project is about turning the familiar story of "the postseason is different" into something more concrete. Instead of just accepting that anything can happen in October, it asks whether there are consistent patterns behind the teams that keep winning series and the ones that go home early. The answers won't remove all the randomness that makes the playoffs

exciting, but they can help explain why certain teams feel built for October—and what might be missing when they fall short.

Research Questions:

1. Which team metrics predict series wins most strongly?
    - Compare hitting metrics (OPS, wOBA, HR/PA), pitching metrics (ERA-, FIP-, K-BB%), and defense (Def runs saved, OAA) vs. round advancement.
2. Is run prevention (pitching + defense) more important than run production in October?
    - Model series outcomes using composite prevention metrics (staff K-BB%, WHIP, bullpen xFIP in high leverage, team OAA) vs. production metrics (wOBA, ISO).
3. How does home-run rate and quality of contact relate to postseason success?
    - Examine HR rate, Barrel% and Hard-Hit% vs. series outcomes.
4. Do deeper bullpens and high-leverage performance drive advancement?
    - Use bullpen innings share, average Leverage Index (gmLI), and late-inning K-BB%/xFIP to predict wins.
5. Is clutch performance (late/close) a reliable predictor or just noise?
    - Compare regular-season “late & close” production to postseason results; test stability year-over-year.
6. How big is home-field advantage in best-of series?
    - Estimate win probability differences by home vs. away, controlling for opponent strength.
7. Does contact quality vs. contact rate matter more in October?
    - Contrast high-contact teams (low K%) with high-damage teams (high ISO/Barrel%) on advancement.

8. Does baserunning (SB%, Extra Bases Taken) correlate with postseason wins?
    - Analyze SB attempts/success and BsR (baserunning runs) vs. round outcomes.
9. Which individual advanced stats (WAR, WPA, Win Probability Added in postseason, xwOBA) best associate with deep runs?
    - Compare profiles of teams with multiple high-WAR or high-WPA performers vs. early exits.
10. Do teams with multiple frontline starters outperform those led by one ace?

- Measure playoff rotation depth (top-3 SP xERA/xFIP) vs. series advancement.

Data Gathering

Direct Data Download(s)

1) Kaggle – Dodgers Batting & Pitching (1958 – 2023)
    a. https://www.kaggle.com/datasets/mattop/los-angeles-dodgers-batting-and-pitching-1958-2023

Rank	Year	Position	Name	Age	Games	Plate_Appe	At_Bats	Runs	Hits	Doubles	Triples	Home_Rur	Runs_Batt	Stolen_Bas	Caught_Sto	Base_On_E	Strikeouts	Batting_Av	On_Base_F	Slugging_P	On_Base_F	On_Base_F	Total_Base	Double_Pla	Times_Hit	Sacrifice_	Sacrifice_F
1	2023	C	Will Smith	28	126	554	464	80	121	21	2	19	76	3	0	63	89	0.261	0.359	0.438	0.797	114	203	8	15	0	12
2	2023	1B	Freddie Fr	33	161	730	637	131	211	59	2	29	102	23	1	72	121	0.331	0.41	0.567	0.976	161	361	14	16	0	5
3	2023	2B	Miguel Var	23	81	304	256	36	50	15	4	7	32	3	2	38	61	0.195	0.305	0.367	0.672	81	94	1	4	1	4
4	2023	SS	Miguel Roj	34	124	423	385	49	91	16	1	5	31	8	3	26	48	0.236	0.29	0.322	0.612	66	124	12	5	2	5
5	2023	3B	Max Muncy	32	135	579	482	95	102	17	1	36	105	1	2	85	153	0.212	0.333	0.475	0.808	115	229	8	6	0	6
6	2023	LF	David Pera	35	133	422	394	47	102	25	1	7	55	4	1	20	72	0.259	0.294	0.381	0.675	81	150	8	2	0	6
7	2023	CF	James Out	26	151	567	483	86	120	16	3	23	70	16	3	68	181	0.248	0.353	0.437	0.79	112	211	4	12	0	4
8	2023	RF	Mookie Be	30	152	693	584	126	179	40	1	39	107	14	3	96	107	0.307	0.408	0.579	0.987	163	338	5	8	0	5
9	2023	DH	J.D. Martin	35	113	479	432	61	117	27	2	33	103	1	0	34	149	0.271	0.321	0.572	0.893	134	247	11	2	0	9
10	2023	UT	Chris Taylc	32	117	384	338	51	80	15	1	15	56	16	3	41	125	0.237	0.326	0.42	0.746	100	142	2	4	0	1
11	2023	RF	Jason Hey	33	124	377	334	56	90	23	0	15	40	2	2	34	64	0.269	0.34	0.473	0.813	117	158	7	3	0	3
12	2023	C	Austin Barı	33	59	200	178	15	32	5	0	2	11	2	1	17	43	0.18	0.256	0.242	0.498	36	43	7	2	1	2
13	2023	UT	Enrique He	31	54	185	168	19	44	12	0	5	30	1	0	12	29	0.262	0.308	0.423	0.731	95	71	6	1	0	4
14	2023	2B	Amed Rosa	27	48	133	125	19	32	6	2	3	18	6	2	7	22	0.256	0.301	0.408	0.709	89	51	4	1	0	0
15	2023	OF	Trayce Tho	32	36	87	71	12	11	0	0	5	14	0	0	15	37	0.155	0.31	0.366	0.677	82	26	0	1	0	0
16	2023	IF	Michael Bu	25	27	81	72	9	12	3	0	2	7	1	0	8	27	0.167	0.247	0.292	0.539	45	21	0	0	0	1
17	2023	OF	Jonny Delu	24	24	45	42	5	11	1	0	2	6	1	0	3	8	0.262	0.311	0.429	0.74	97	18	0	0	0	0
18	2023	2B	Kolten Wor	32	20	34	30	4	9	0	0	2	8	2	1	2	7	0.3	0.353	0.5	0.853	127	15	1	1	0	1
19	2023	IF	Yonny Hern	25	14	27	22	5	3	1	0	0	4	0	0	2	8	0.136	0.231	0.182	0.413	14	4	0	1	1	1
20	2023	C	Austin Wyr	32	5	12	11	0	2	1	0	0	2	0	0	1	5	0.182	0.25	0.273	0.523	42	3	0	0	0	0
21	2023	SS	Luke Willia	26	4	10	10	0	1	0	0	0	0	1	0	0	3	0.1	0.1	0.1	0.2	-46	1	0	0	0	0
22	2023	CF	Jake Maris	32	4	6	5	0	2	0	0	0	0	0	1	0	0	0.4	0.5	0.4	0.9	149	2	0	1	0	0
23	2023	P	Shelby Mill	32	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-100	0	0	0	0	0
24	2023	P	Michael Gr	26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
25	2023	P	Dustin May	25	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
26	2023	P	Noah Synd	30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 2: Player Stats w/ advanced metrics from Excel

The dataset I'm using comes from Kaggle and contains team-season batting and pitching data for the Los Angeles Dodgers from 1958 through 2023. It's a labeled, quantitative time series that spans the club's modern era in Los Angeles and is well-suited for long-form content. Furthermore, the Dodgers are a consistent winning team which allows for data to explore why exactly the Dodgers are such an incredible franchise.

On the batting side, the file typically includes identifiers such as season/year along with common team metrics—for example G/PA/AB (games, plate appearances, at-bats), R/H/2B/3B/HR/RBI, BB/SO/SB/CS, and rate stats like AVG (batting average), OBP (on-base percentage), SLG (slugging), and OPS (on-base plus slugging). The pitching portion provides G/GS/IP (games, starts, innings), H/R/ER/HR, BB/SO, and run-prevention/efficiency indicators such as ERA and often WHIP; many historical team datasets also include saves and other summary totals. Together, these variables give a comprehensive view of the Dodgers' offensive

production and run prevention by season, enabling comparisons across the time periods of Dodgers dominance.

This dataset is valuable for studying postseason readiness and overall team quality because it allows you to connect run creation (e.g., OBP/SLG/OPS, HR rate) with run prevention (e.g., ERA, WHIP, strikeout-to-walk profile) at the team level. You can, for example, identify seasons characterized by elite contact/OBP teams versus true-outcome power teams, or rotations/bullpens that drove success via high strikeout rates and low walk rates. Since the data are organized by year, it's straightforward to align regular-season performance with postseason outcomes, highlight standout seasons, and quantify how shifts in strategy (small ball vs. power; starter-heavy vs. bullpen-centric) correlate with wins.

2) Kaggle – MLB Pitching Data

   a. https://www.kaggle.com/datasets/josephvm/mlb-game-data?select=games.csv

lum	Pitch	Type	MPH	play-hitzon	play-bases	play-field	Pitcher	Pitching Te	Batting Tea	Inning	Event Id	Game
1	Ball	Fastball	90	top: 14px; right: 13px;			Liriano	PIT	STL	Top 1st	1	3.6E+08
2	Ground Ou	Fastball	90	top: 10.37px; right: 26		top: 16.02	Liriano	PIT	STL	Top 1st	1	3.6E+08
1	Ground Ou	Fastball	90	top: 12.97px; right: 22		top: 17.88	Liriano	PIT	STL	Top 1st	2	3.6E+08
1	Strike Look	Fastball	90	top: 9.5px; right: 28.83px;			Liriano	PIT	STL	Top 1st	3	3.6E+08
2	Ball	Slider	83	top: 22px; right: 23.13px;			Liriano	PIT	STL	Top 1st	3	3.6E+08
3	Ball	Changeup	84	top: 10.59px; right: 30.87px;			Liriano	PIT	STL	Top 1st	3	3.6E+08
4	Ball	Fastball	90	top: 19.47px; right: 30.87px;			Liriano	PIT	STL	Top 1st	3	3.6E+08
5	Foul Ball	Fastball	91	top: 15.57px; right: 25.98px;			Liriano	PIT	STL	Top 1st	3	3.6E+08
6	Strike Look	Fastball	92	top: 16.01px; right: 17.02px;			Liriano	PIT	STL	Top 1st	3	3.6E+08
1	Strike Look	Fastball	90	top: 14.05px; right: 19.87px;			Wainwrigh	STL	PIT	Bottom 1st	5	3.6E+08
2	Ground Ou	Slider	88	top: 9.72px; right: 23.1		top: 15.79	Wainwrigh	STL	PIT	Bottom 1st	5	3.6E+08
1	Strike Look	Fastball	90	top: 14.92px; right: 27.61px;			Wainwrigh	STL	PIT	Bottom 1st	6	3.6E+08
2	Ball	Fastball	90	top: 22px; right: 12.33px;			Wainwrigh	STL	PIT	Bottom 1st	6	3.6E+08
3	Foul Ball	Curve	73	top: 14.05px; right: 21.3px;			Wainwrigh	STL	PIT	Bottom 1st	6	3.6E+08
4	Hit By Pitcl	Slider	85	top: 0px; ri	1		Wainwrigh	STL	PIT	Bottom 1st	6	3.6E+08
1	Ball	Fastball	89	top: 18.17	1		Wainwrigh	STL	PIT	Bottom 1st	7	3.6E+08
2	Strike Look	Slider	86	top: 10.15px; right: 25.37px;			Wainwrigh	STL	PIT	Bottom 1st	7	3.6E+08
3	Ball	Curve	74	top: 22px; right: 31.69px;			Wainwrigh	STL	PIT	Bottom 1st	7	3.6E+08
4	Foul Ball	Slider	87	top: 7.12px; right: 20.07px;			Wainwrigh	STL	PIT	Bottom 1st	7	3.6E+08
5	Ball	Curve	76	top: 22px; right: 36.17px;			Wainwrigh	STL	PIT	Bottom 1st	7	3.6E+08
6	Single	Fastball	90	top: 7.55p	13	top: 6.86p	Wainwrigh	STL	PIT	Bottom 1st	7	3.6E+08
1	Ball	Slider	87	top: 14.92	13		Wainwrigh	STL	PIT	Bottom 1st	8	3.6E+08
2	Line Out	Fastball	89	top: 7.99p	13	top: 14.52	Wainwrigh	STL	PIT	Bottom 1st	8	3.6E+08
1	Strike Swin	Slider	87	top: 19.69	13		Wainwrigh	STL	PIT	Bottom 1st	9	3.6E+08
2	Strike Look	Slider	88	top: 13.84px; right: 25.37px;			Wainwrigh	STL	PIT	Bottom 1st	9	3.6E+08

Figure 3: Pitching Data

The data-set I've gathered is from the website Kaggle. It is a mixed data set that contains both qualitative and quantitative data sets. The qualitative columns are the pitcher, pitch, type of pitch, pitching team, batting team and inning. The qualitative columns are the mph (miles per hour), the play-hitzone, play-bases and play-field. This is a comprehensive dataset for pitching that allows to see what specific pitches are strongest against specific teams. It goes through each individual pitch for every team. This creates an incredibly large dataset that was too massive to see past a certain point. There are over 8 columns (variables) and has over 1000 rows. In order to see everything, I must use python to extract the specific content that I want to use.

This data-set will help forward me in my goal to assert what specific pitches work directly towards play-off success. What's the average speed for pitches to obtain maximum success? What percentage of pitches work better from the 1st Inning to the 9th Inning? Are there any types of pitches that work better during overtime? All these questions can be answered through the amount of extensive pitching data from the data-set.

Data Gathering Using an API

1) MLB StatsAPI (No Key Required)
    a. Base URL: https://statsapi.mlb.com/api/v1
    b. Query URL: https://statsapi.mlb.com/api/v1/schedule/postseason?season=2024

c. Box Scores: https://statsapi.mlb.com/api/v1/game/{gamePk}/boxscore

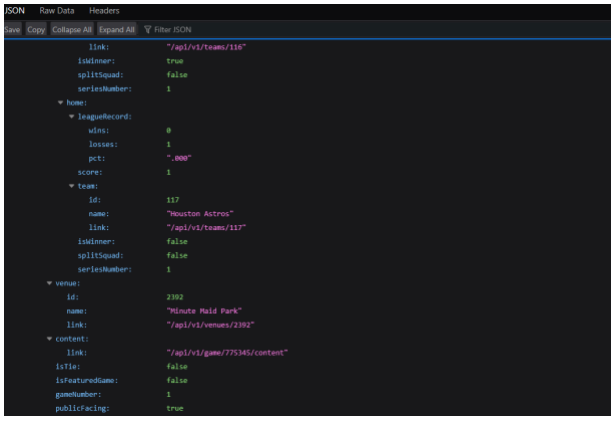


Figure 4: JSON Data

I originally chose to work on the NBA play-off's but obtaining an API for NBA stats was too complicated and expensive. I used the MLB Stats API because they do not require a key nor is it expensive. Everything is public free information which allows the actual data analysis to be easier. It's an official JSON API that exposes teams, schedules, games, box-scores, play-by-play,

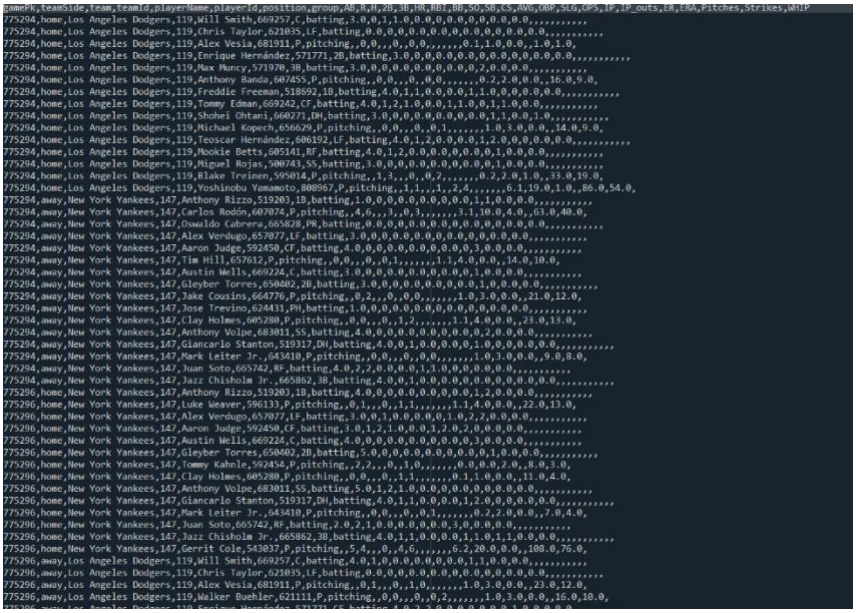


Figure 5: MLB Postseason 2024 Raw Data from Python

and advanced stats. I created a python code to extract specific data from a certain date during the MLB season. I chose to extract the postseason stats during the 2024 Play-Off's.

With my code, I requested a specific date – the postseason games for the 2024 season. I called upon the postseason bracket endpoint: *GET /schedule/postseason?season=2024*. Next, I traversed the nested JSON and collected every gamePK (Game ID) value. This allowed for me to obtain every postseason game for 2024. For every gamePK, I called *GET /game/{gamePK}/*boxscore. This allowed me to obtain data for each player on each team. The specific stats I got were for both sides, Batting and Pitching. The specific stats are listed below.

Batting

- AB (At-Bats)
- R (Runs Scored)
- H (Hits)
- 2B (Doubles)
- 3B (Triples)
- HR (Home Runs)
- RBI (Runs Batted In)
- BB (Walks / Bases on Balls)
- SO (Strikeouts)
- SB (Stolen Bases)
- CS (Caught Stealing)

- AVG (Batting Average)
- OBP (On-Base Percentage)
- SLG (Slugging Percentage)
- OPS (On-Base Plus Slugging)

Pitching

- IP (Innings Pitched — measured in outs/3; e.g., 5.2 = 5 innings + 2 outs)
- H (Hits Allowed)
- R (Runs Allowed)
- ER (Earned Runs)
- BB (Walks Issued / Bases on Balls)
- SO (Strikeouts)
- HR (Home Runs Allowed)
- ERA (Earned Run Average — earned runs per 9 IP)
- Pitches (Total Pitches Thrown)
- Strikes (Total Strikes Thrown)
- WHIP (Walks + Hits per Inning Pitched)

After obtaining the data, I normalized the JSON into a flat table with one row per player-game. Upon the finished data, I saved everything to CSV file for data analysis.

The data-set will allow me to reach my goal of specific traits to look out for during the play-offs. What type of players step up the most during the postseason? Is there a specific type of player that is necessary?

Data Cleaning

Los Angeles Dodgers Regular Season Batting Data

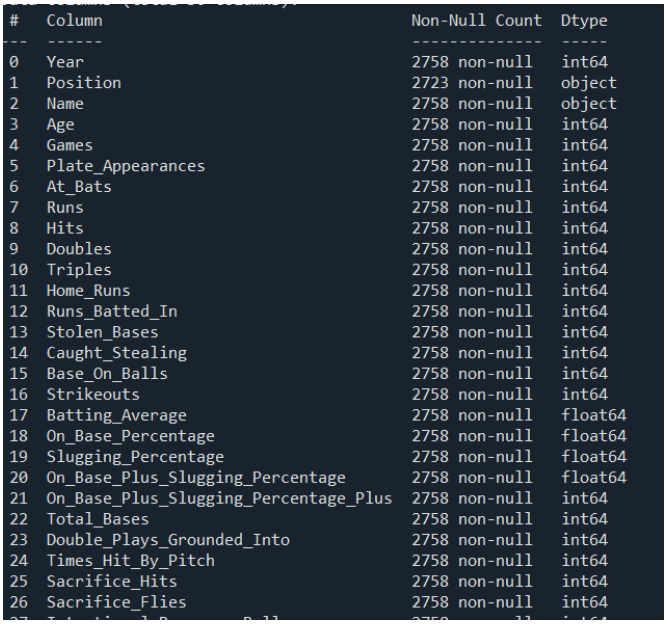
Rank	Year	Position	Name	Age	Games	Plate_Appe	At_Bats	Runs	Hits	Doubles	Triples	Home_Rur	Runs_Batte	Stolen_Bas	Caught_Ste	Base_On_E	Strikeouts	Batting_Av	On_Base_F	Slugging_P	On_Base_F	On_Base_F	Total_Bas
1	2023	C	Will Smith	28	126	554	464	80	121	21	2	19	76	3	0	63	89	0.261	0.359	0.438	0.797	114	20
2	2023	1B	Freddie Fre	33	161	730	637	131	211	59	2	29	102	23	1	72	121	0.331	0.41	0.567	0.976	161	36
3	2023	2B	Miguel Var	23	81	304	256	36	50	15	4	7	32	3	2	38	61	0.195	0.305	0.367	0.672	81	9
4	2023	SS	Miguel Roj	34	124	423	385	49	91	16	1	5	31	8	3	26	48	0.236	0.29	0.322	0.612	66	12
5	2023	3B	Max Muncy	32	135	579	482	95	102	17	1	36	105	1	2	85	153	0.212	0.333	0.475	0.808	115	22
6	2023	LF	David Pera	35	133	422	394	47	102	25	1	7	55	4	1	20	72	0.259	0.294	0.381	0.675	81	15
7	2023	CF	James Out	26	151	567	483	86	120	16	3	23	70	16	3	68	181	0.248	0.353	0.437	0.79	112	21
8	2023	RF	Mookie Bet	30	152	693	584	126	179	40	1	39	107	14	3	96	107	0.307	0.408	0.579	0.987	163	33
9	2023	DH	J.D. Martin	35	113	479	432	61	117	27	2	33	103	1	0	34	149	0.271	0.321	0.572	0.893	134	24
10	2023	UT	Chris Taylo	32	117	384	338	51	80	15	1	15	56	16	3	41	125	0.237	0.326	0.42	0.746	100	14
11	2023	RF	Jason Heyw	33	124	377	334	56	90	23	0	15	40	2	2	34	64	0.269	0.34	0.473	0.813	117	15
12	2023	C	Austin Barn	33	59	200	178	15	32	5	0	2	11	2	1	17	43	0.18	0.256	0.242	0.498	36	4
13	2023	UT	Enrique He	31	54	185	168	19	44	12	0	5	30	1	0	12	29	0.262	0.308	0.423	0.731	95	7
14	2023	2B	Amed Rosa	27	48	133	125	19	32	6	2	3	18	6	2	7	22	0.256	0.301	0.408	0.709	89	5
15	2023	OF	Trayce Tho	32	36	87	71	12	11	0	0	5	14	0	0	15	37	0.155	0.31	0.366	0.677	82	2
16	2023	IF	Michael Bu	25	27	81	72	9	12	3	0	2	7	1	0	8	27	0.167	0.247	0.292	0.539	45	2
17	2023	OF	Jonny Delu	24	24	45	42	5	11	1	0	2	6	1	0	3	8	0.262	0.311	0.429	0.74	97	1
18	2023	2B	Kolten Wor	32	20	34	30	4	9	0	0	2	8	2	1	2	7	0.3	0.353	0.5	0.853	127	1
19	2023	IF	Yonny Hern	25	14	27	22	5	3	1	0	0	4	0	0	2	8	0.136	0.231	0.182	0.413	14	
20	2023	C	Austin Wyr	32	5	12	11	0	2	1	0	0	2	0	0	1	5	0.182	0.25	0.273	0.523	42	
21	2023	SS	Luke Willia	26	4	10	10	0	1	0	0	0	0	1	0	0	3	0.1	0.1	0.1	0.2	-46	
22	2023	CF	Jake Maris	32	4	6	5	0	2	0	0	0	0	0	1	0	0	0.4	0.5	0.4	0.9	149	
23	2023	P	Shelby Mill	32	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-100	
24	2023	P	Michael Gr	26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
25	2023	P	Dustin May	25	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
26	2023	P	Noah Synd	30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
27	2023	P	Jake Reed	30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
28	2023	P	Julio UrÃ-a	26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Figure 6: Before Batting Data for Dodgers

I am using the Los Angeles Dodgers batting dataset and will perform targeted cleaning. First, I used Python to validate data integrity by inspecting each column's data type; all fields matched their expected types. Rather than correcting types, I will streamline the dataset by removing variables that are not needed for my analysis. The retained columns are **Year**, **Position**, **Name**, **Age**, **Batting Average**, **On-Base Percentage**, and **Slugging Percentage**. Their data types are as follows: Year — *int64*; Position — *object*; Name — *object*; Age — *int64*; Batting Average — *float64*; On-Base Percentage — *float64*; Slugging Percentage — *float64*. These results indicate no type-related quality issues. For example, if year had a word in it, it would no longer say that it was an *int64*. The same could be said in reverse for columns such as Batting Average or On-Base Percentage.

I will use **Year**, **Position**, **Name**, and **Age** to identify each player and provide their respective role and developmental context. **At-Bat** quantifies how many times a player was given a chance to bat. Batting **Average** is included as a core measure of contact and hit frequency, while **On-Base Percentage** captures the ability to reach base (including walks and HBP), and **Slugging Percentage** reflects power production. Together, these variables offer a concise, non-redundant view of player performance suitable for trend analysis and comparative evaluation.

To ensure that data is completely clean, I used Python to see if there were any empty columns. I found out there are no empty columns.



#	Column	Non-Null Count	Dtype
0	Year	2758 non-null	int64
1	Position	2723 non-null	object
2	Name	2758 non-null	object
3	Age	2758 non-null	int64
4	Games	2758 non-null	int64
5	Plate_Appearances	2758 non-null	int64
6	At_Bats	2758 non-null	int64
7	Runs	2758 non-null	int64
8	Hits	2758 non-null	int64
9	Doubles	2758 non-null	int64
10	Triples	2758 non-null	int64
11	Home_Runs	2758 non-null	int64
12	Runs_Batted_In	2758 non-null	int64
13	Stolen_Bases	2758 non-null	int64
14	Caught_Stealing	2758 non-null	int64
15	Base_On_Balls	2758 non-null	int64
16	Strikeouts	2758 non-null	int64
17	Batting_Average	2758 non-null	float64
18	On_Base_Percentage	2758 non-null	float64
19	Slugging_Percentage	2758 non-null	float64
20	On_Base_Plus_Slugging_Percentage	2758 non-null	float64
21	On_Base_Plus_Slugging_Percentage_Plus	2758 non-null	int64
22	Total_Bases	2758 non-null	int64
23	Double_Plays_Grounded_Into	2758 non-null	int64
24	Times_Hit_By_Pitch	2758 non-null	int64
25	Sacrifice_Hits	2758 non-null	int64
26	Sacrifice_Flies	2758 non-null	int64

Figure 7: Null Count

The next step is to ensure that no data goes beyond the reasonable expectations of stats. For example, **Years** should not be before the 1900s, **Age** should not go below 16 or above 60, **Batting Average** should not go above 1, **On-Base Percentage** should not go above 1.2 and **Slugging Percentage** should not go above 2.5.

```
Bad counts (year, age, BA, OBP, SLG): 0 0 0 0 0
```

Figure 8: Bad Count is 0

The next step is to ensure if there are any outliers for the data. Using Python, I created a boxplot to show the outliers for the **Batting Average**, **On-Base Percentage**, and **Slugging Percentage** to reveal what they are and if they are valid.

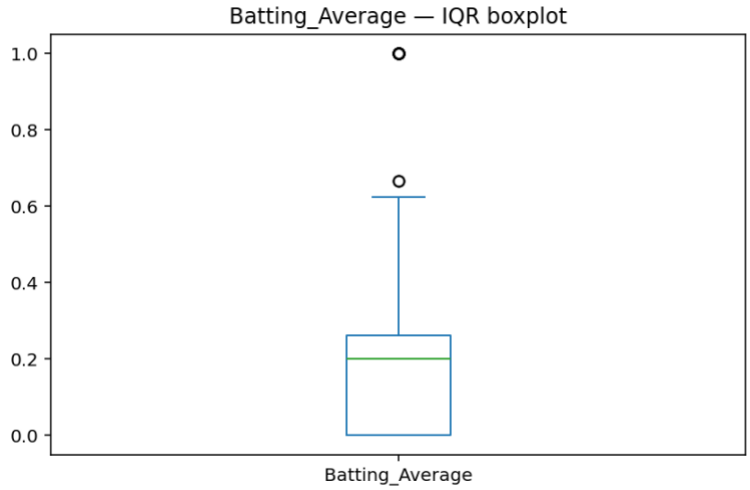
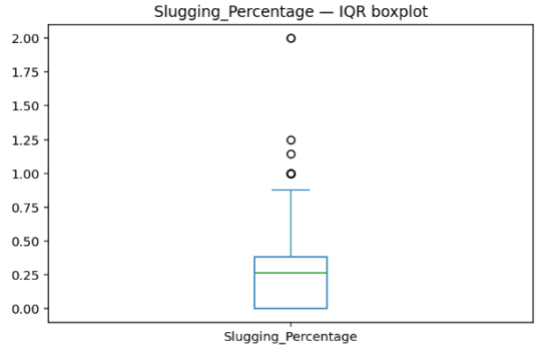
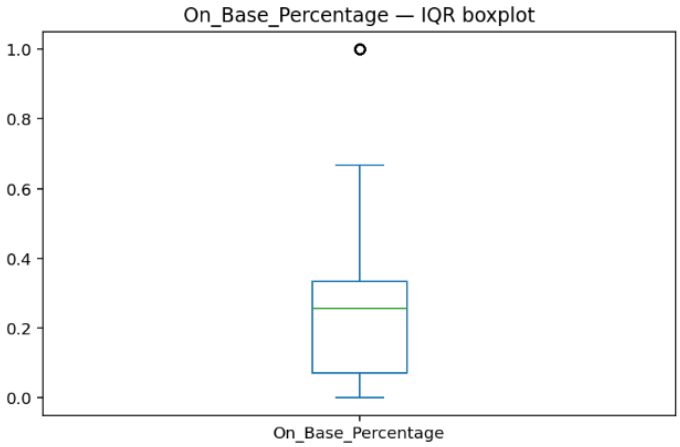


Figure 9: Outliers

Batting Average has a few outliers beyond the medium but can be explained through short sampling of a few players throughout the Dodger's season. They may reflect a player with a few hits over very few at-bats which can post a Batting Average close to 1.0000. With that in mind, there is no need to remove the data from our research.

In fact, the same can be explained for **Slugging Percentage** and **On-Base Percentage**.





A player can have close to near perfection for On Base Percentages and Slugging Percentages with very few at-bats. With that in mind, none of the outliers are necessary to remove. If anything, we can choose to ignore those data variables considering how they will not play a significant role in determining what quantifies a successful player due to the context of the low amounts of at-bats.

Next, the step is to ensure that the data is normalized using min-max. We will only want to use the quantitative columns which is revealed below.

```
=== BEFORE (first 10) ===
   Age  Batting_Average  On_Base_Percentage  Slugging_Percentage  At_Bats
0   28            0.261               0.359                0.438      464
1   33            0.331               0.410                0.567      637
2   23            0.195               0.305                0.367      256
3   34            0.236               0.290                0.322      385
4   32            0.212               0.333                0.475      482
5   35            0.259               0.294                0.381      394
6   26            0.248               0.353                0.437      483
7   30            0.307               0.408                0.579      584
8   35            0.271               0.321                0.572      432
9   32            0.237               0.326                0.420      338
```

Figure 10: Quantitative Columns

Above are the columns that will be used before we begin to use Python to normalize them. After normalizing the variables, the dataset will begin to go between 0 or 1.

```
=== AFTER Min-Max (first 10) ===
        Age  Batting_Average  On_Base_Percentage  Slugging_Percentage   At_Bats
0  0.34375            0.261               0.359               0.2190  0.667626
1  0.50000            0.331               0.410               0.2835  0.916547
2  0.18750            0.195               0.305               0.1835  0.368345
3  0.53125            0.236               0.290               0.1610  0.553957
4  0.46875            0.212               0.333               0.2375  0.693525
5  0.56250            0.259               0.294               0.1905  0.566906
6  0.28125            0.248               0.353               0.2185  0.694964
7  0.40625            0.307               0.408               0.2895  0.840288
8  0.56250            0.271               0.321               0.2860  0.621583
9  0.46875            0.237               0.326               0.2100  0.486331
```

Figure 11: Normalized Min-Max Dataset

The normalized dataset is listed above.

The next step is to create a data frame that contains only unlabeled and quantitative data. Through Python, we created a dataframe of just that for future KMeans analysis.

```
0  2023   28            0.261               0.359               0.438      464
1  2023   33            0.331               0.410               0.567      637
2  2023   23            0.195               0.305               0.367      256
3  2023   34            0.236               0.290               0.322      385
4  2023   32            0.212               0.333               0.475      482
5  2023   35            0.259               0.294               0.381      394
6  2023   26            0.248               0.353               0.437      483
7  2023   30            0.307               0.408               0.579      584
8  2023   35            0.271               0.321               0.572      432
9  2023   32            0.237               0.326               0.420      338
```

Figure 12: Quantitative Dataframe

Overall, the Dodger's Batting Records are now secure and ready to use.

Year	Position	Name	Age	Batting_Av	On_Base_F	Slugging_P	At_Bats
2023	C	Will Smith	28	0.261	0.359	0.438	464
2023	1B	Freddie Fr	33	0.331	0.41	0.567	637
2023	2B	Miguel Var	23	0.195	0.305	0.367	256
2023	SS	Miguel Roj	34	0.236	0.29	0.322	385
2023	3B	Max Muncy	32	0.212	0.333	0.475	482
2023	LF	David Pera	35	0.259	0.294	0.381	394
2023	CF	James Out	26	0.248	0.353	0.437	483
2023	RF	Mookie Be	30	0.307	0.408	0.579	584
2023	DH	J.D. Martin	35	0.271	0.321	0.572	432
2023	UT	Chris Taylo	32	0.237	0.326	0.42	338
2023	RF	Jason Hey	33	0.269	0.34	0.473	334
2023	C	Austin Barı	33	0.18	0.256	0.242	178
2023	UT	Enrique He	31	0.262	0.308	0.423	168
2023	2B	Amed Rosa	27	0.256	0.301	0.408	125
2023	OF	Trayce Tho	32	0.155	0.31	0.366	71
2023	IF	Michael Bu	25	0.167	0.247	0.292	72
2023	OF	Jonny Delu	24	0.262	0.311	0.429	42
2023	2B	Kolten Wor	32	0.3	0.353	0.5	30
2023	IF	Yonny Hern	25	0.136	0.231	0.182	22
2023	C	Austin Wyr	32	0.182	0.25	0.273	11
2023	SS	Luke Willia	26	0.1	0.1	0.1	10
2023	CF	Jake Maris	32	0.4	0.5	0.4	5
2023	P	Shelby Mill	32	0	0	0	1

Figure 13: After Clean Dataset

Los Angeles Dodgers Batting Post-Season Stats

I will be using dataset that I received from the MLB API. I will be reviewing information about the Dodger's Post-Season Stats for Batting. I have a clean data-set from the MLB API which gave me the results that I wanted.

A	B	C	D	E	F	G	H
season	playerName	primaryPo	Age	AB	AVG	OBP	SLG
1995	Antonio Osuna	P	22	0	0	0	0
1995	Billy Ashley	PH	25	0	0	1	0
1995	Brett Butler	CF	38	15	0.266667	0.266667	0.266667
1995	Chad Fonville	SS	24	12	0.5	0.5	0.5
1995	Chris Gwynn	PH	30	1	0	0	0
1995	Dave Hansen	PH	26	3	0.666667	0.666667	0.666667
1995	Delino DeShields	2B	26	12	0.25	0.307692	0.25
1995	Eric Karros	1B	27	12	0.5	0.538462	1.083333
1995	Hideo Nomo	P	27	2	0	0	0
1995	Ismael Valdez	P	22	3	0	0	0
1995	John Cummings	P	26	0	0	0	0
1995	Jose Offerman	PR	26	0	0	0	0
1995	Kevin Tapani	P	31	0	0	0	0
1995	Mark Guthrie	P	30	0	0	0	0
1995	Mike Piazza	C	27	14	0.214286	0.214286	0.5
1995	Mitch Webster	PH	36	2	0	0	0
1995	Pedro Astacio	P	26	0	0	0	0
1995	Ramon Martinez	P	27	1	0	0	0
1995	Raul Mondesi	RF	24	9	0.222222	0.3	0.222222
1995	Roberto Kelly	LF	31	11	0.363636	0.416667	0.363636
1995	Tim Wallach	3B	38	12	0.083333	0.153846	0.083333
1995	Todd Hollandsworth	RF	22	2	0	0	0
1996	Antonio Osuna	P	23	0	0	0	0
1996	Billy Ashley	PH	26	2	0	0	0

Figure 14: Before Data

The data I have displayed are the **post-season year**, **playerName**, **primaryPosition**, **Age**, **AB**, **AVG**, **OBP**, and **SLG**. One way I was able to verify the results were all good was the data type. . Their data types are as follows: Year — *int64*; Position — *object*; Name — *object*; Age — *int64*; Batting Average — *float64*; On-Base Percentage — *float64*; Slugging Percentage — *float64 and* At-Bats – *float64*. These results indicate no type-related quality issues. For example, if year had

a word in it, it would no longer say that it was an *int64*. The same could be said in reverse for columns such as Batting Average or On-Base Percentage.

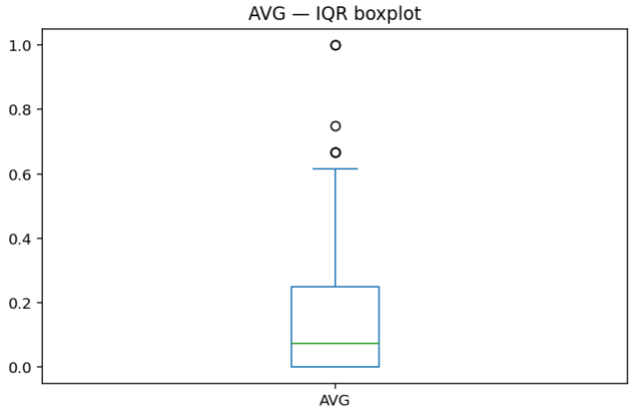
I will use **Year**, **Position**, **Name**, and **Age** to identify each player and provide their respective role and developmental context. **At-Bat** quantifies how many times a player was given a chance to bat. Batting **Average** is included as a core measure of contact and hit frequency, while **On-Base Percentage** captures the ability to reach base (including walks and HBP), and **Slugging Percentage** reflects power production. Together, these variables offer a concise, non-redundant view of player performance suitable for trend analysis and comparative evaluation. The main difference between the previous data sample of the Dodgers was that the previous dataset encompasses the regular season while the dataset I have for this encompasses only the postseason. We can then deduce the difference between a regular season player and a postseason player.

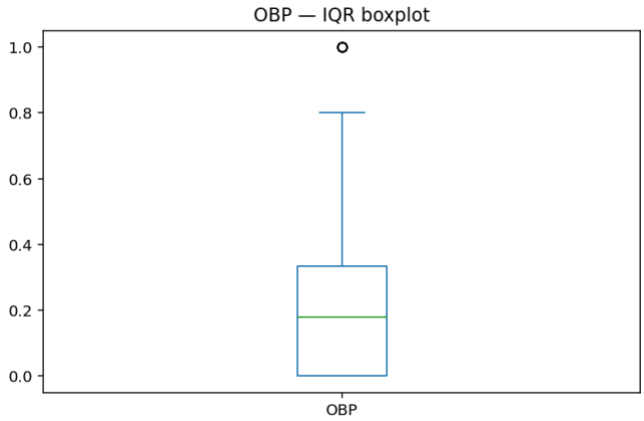
```
Data columns (total 8 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   season           406 non-null    int64
 1   playerName       406 non-null    object
 2   primaryPosition  406 non-null    object
 3   Age              406 non-null    int64
 4   AB               406 non-null    int64
 5   AVG              406 non-null    float64
 6   OBP              406 non-null    float64
 7   SLG              406 non-null    float64
```

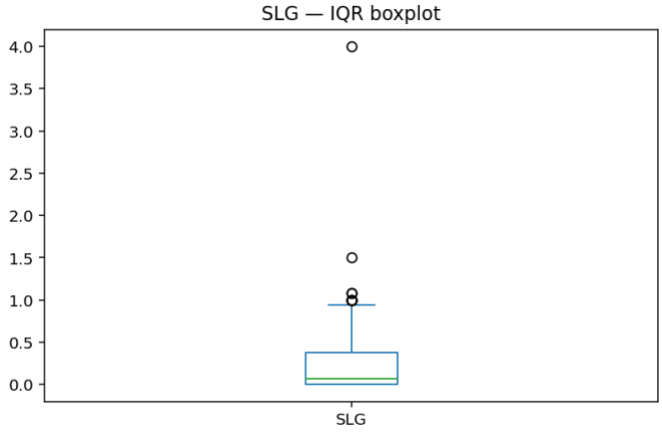
Figure 15: DataTypes and Non-Null Count

Using python, we were able to realize that there were no missing values for the data. There was a non-null count which meant there were no missing values.

The next step is to ensure that there are no outliers in the data. Using a boxplot for each data column, we can determine what the outliers are and if they are a valid target or not.







The box-plots all reveal that there were outliers for each position. However, we can determine that the values are not exaggerative nor are they invalid. Looking at the At-Bat's for each player, we can determine that the players that had a near 100% rate for Batting Average, On-Base Percentage and Slugging Percentage was due to the fact that they had an usually low amount of at-bats which allowed them to have such a high rate.

The next step is to create a data frame so that the new data frame is normalized using min-max. Using python, we can show the difference between the original dataset and the after-normalization dataset.

```
=== BEFORE (first 10) ===
   Age  AB       AVG       OBP       SLG
0   22   0  0.000000  0.000000  0.000000
1   25   0  0.000000  1.000000  0.000000
2   38  15  0.266667  0.266667  0.266667
3   24  12  0.500000  0.500000  0.500000
4   30   1  0.000000  0.000000  0.000000
5   26   3  0.666667  0.666667  0.666667
6   26  12  0.250000  0.307692  0.250000
7   27  12  0.500000  0.538462  1.083333
8   27   2  0.000000  0.000000  0.000000
9   22   3  0.000000  0.000000  0.000000
```

Figure 16: Original Dataset

```
=== AFTER Min-Max (first 10) ===
        Age        AB       AVG       OBP       SLG
0  0.090909  0.000000  0.000000  0.000000  0.000000
1  0.227273  0.000000  0.000000  1.000000  0.000000
2  0.818182  0.211268  0.266667  0.266667  0.066667
3  0.181818  0.169014  0.500000  0.500000  0.125000
4  0.454545  0.014085  0.000000  0.000000  0.000000
5  0.272727  0.042254  0.666667  0.666667  0.166667
6  0.272727  0.169014  0.250000  0.307692  0.062500
7  0.318182  0.169014  0.500000  0.538462  0.270833
8  0.318182  0.028169  0.000000  0.000000  0.000000
9  0.090909  0.042254  0.000000  0.000000  0.000000
```

Figure 17: Normalization Dataset

Next, we will create a new data frame that contains only unlabeled and quantitative data to use for Kmeans. We can use python to build that.

```
0    1995  0.090909  0.000000  0.000000  0.000000  0.000000
1    1995  0.227273  0.000000  0.000000  1.000000  0.000000
2    1995  0.818182  0.211268  0.266667  0.266667  0.066667
3    1995  0.181818  0.169014  0.500000  0.500000  0.125000
4    1995  0.454545  0.014085  0.000000  0.000000  0.000000
5    1995  0.272727  0.042254  0.666667  0.666667  0.166667
6    1995  0.272727  0.169014  0.250000  0.307692  0.062500
7    1995  0.318182  0.169014  0.500000  0.538462  0.270833
8    1995  0.318182  0.028169  0.000000  0.000000  0.000000
9    1995  0.090909  0.042254  0.000000  0.000000  0.000000
```

Figure 18: Kmeans Dataset

Overall, the dataset is clean as it is already as it is taken from the MLB API which is a direct download from the MLB website. Now we can use the data to compare the different stats between the regular season vs the post-season.

	A	B	C	D	E	F	G	H
	season	playerName	primaryPo	Age	AB	AVG	OBP	SLG
2	1995	Antonio Osuna	P	22	0	0	0	0
3	1995	Billy Ashley	PH	25	0	0	1	0
4	1995	Brett Butler	CF	38	15	0.266667	0.266667	0.266667
5	1995	Chad Fonville	SS	24	12	0.5	0.5	0.5
6	1995	Chris Gwynn	PH	30	1	0	0	0
7	1995	Dave Hansen	PH	26	3	0.666667	0.666667	0.666667
8	1995	Delino DeShields	2B	26	12	0.25	0.307692	0.25
9	1995	Eric Karros	1B	27	12	0.5	0.538462	1.083333
10	1995	Hideo Nomo	P	27	2	0	0	0
11	1995	Ismael Valdez	P	22	3	0	0	0
12	1995	John Cummings	P	26	0	0	0	0
13	1995	Jose Offerman	PR	26	0	0	0	0
14	1995	Kevin Tapani	P	31	0	0	0	0
15	1995	Mark Guthrie	P	30	0	0	0	0
16	1995	Mike Piazza	C	27	14	0.214286	0.214286	0.5
17	1995	Mitch Webster	PH	36	2	0	0	0
18	1995	Pedro Astacio	P	26	0	0	0	0

Figure 19: Clean Dataset

Unsupervised Learning with Kmeans Clustering

Postseason Dodger Batter Stats

I plan to use the dataset for the postseason stats for the Dodgers batters. I already had the data clean earlier for K means which means we're ready to visualize the data.

```
=== Quantitative-only dataframe for KMeans (first 10) ===
   season       Age        AB       AVG       OBP       SLG
0    1995  0.090909  0.000000  0.000000  0.000000  0.000000
1    1995  0.227273  0.000000  0.000000  1.000000  0.000000
2    1995  0.818182  0.211268  0.266667  0.266667  0.066667
3    1995  0.181818  0.169014  0.500000  0.500000  0.125000
4    1995  0.454545  0.014085  0.000000  0.000000  0.000000
5    1995  0.272727  0.042254  0.666667  0.666667  0.166667
6    1995  0.272727  0.169014  0.250000  0.307692  0.062500
7    1995  0.318182  0.169014  0.500000  0.538462  0.270833
8    1995  0.318182  0.028169  0.000000  0.000000  0.000000
9    1995  0.090909  0.042254  0.000000  0.000000  0.000000
Shape (rows, features): (406, 6)
```

Figure 20: Clean Dataframe

We prepared the dataset earlier by removing all quantitative columns while keeping the quantitative columns.

Visualizing the data is a key method to understanding the data. As only three columns can be used, I will pick AB (At Bat), OBP (On-Base Percentage) and SLG (Slugging Percentage). I picked those 3 variables as it will allow us to fully understand how strong the Dodgers postseason battings are for postseason success. We will remove the other columns to make sure we can begin K means analysis.

```
         AB       OBP       SLG
0  0.000000  0.000000  0.000000
1  0.000000  1.000000  0.000000
2  0.211268  0.266667  0.066667
3  0.169014  0.500000  0.125000
4  0.014085  0.000000  0.000000
5  0.042254  0.666667  0.166667
6  0.169014  0.307692  0.062500
7  0.169014  0.538462  0.270833
8  0.028169  0.000000  0.000000
9  0.042254  0.000000  0.000000
Shape (rows, features): (406, 3)
```

Figure 21: K means Data

Now that we have our finished dataset, we will begin to visualize the code with a 3D visualization.

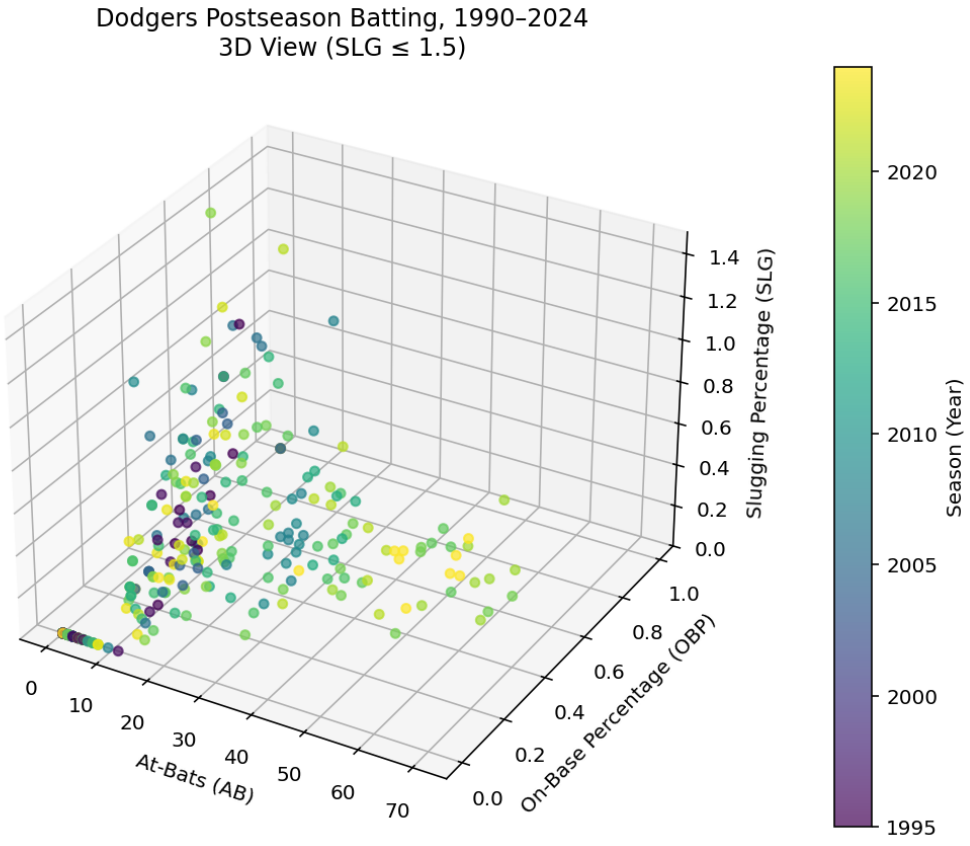


Figure 22: At-Bats vs On Base Percentage vs Slugging Percentage

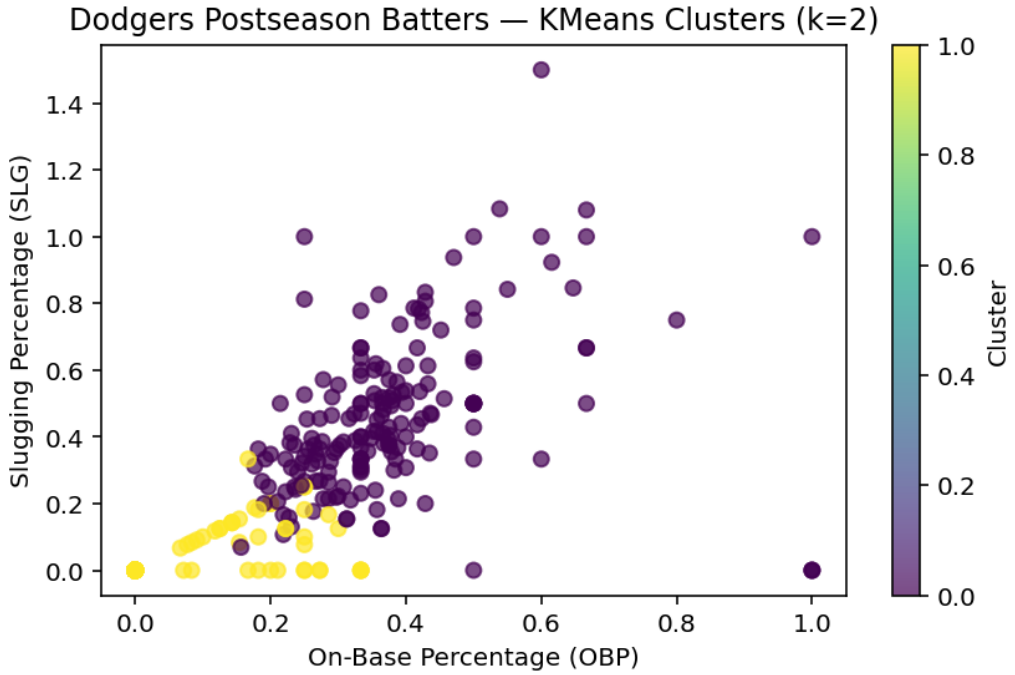
In this 3D scatterplot, each point represents a Dodgers hitter in a given postseason. The three axes show At-Bats, On-Base Percentages and Slugging Percentages, while the color of each point reveals the season, with darker colors for older years and brighter yellow for more recent seasons. Most of the players fall with a dense cluster between 5-35 At-Bats, OBP from roughly 0.2-0.6 and SLG between 0.2-0.9 range. The graph reveals that the Dodgers, as a postseason team seem to perform relatively well on a normal basis.

For this view, I limited SLG to values of 1.5 and below which removed a few extreme small-sample outliers. Thios allows it to make it much easier to see the main cluster of typical performances for the players. Within the main cluster, there is a visible positive relationship between OBP and SLG: hitters who reach base more often also tend to hit for more power; nonetheless, there is still a noticeable spread that reveals how some players contribute more through getting on base rather than contributing more through slugging.

Looking at the colors and the extremes on each axis reveals even more patterns. Along the right side of the plot, there are a cluster of players with higher At-Bats (around 60-70) that are more recent (2015 and up). This reveals that there have been more regulars who have accumulated a lot of postseason opportunities for the Dodgers. This lines up with the Dodgers becoming more and more successful recently than they were decades ago.

The next step is to apply k means clustering to the prepared dataset. We will use the library from Sklearn to discuss the K means using k = 2, k = 3 and k = 4.

K = 2



For the value of k = 2, we have split the hitters into 2 groups based on AB, OBP and SLG.

```
Cluster centers (min-max SCALED AB, OBP, SLG):
      AB    OBP    SLG
0  0.347  0.375  0.303
1  0.034  0.036  0.012

Cluster centers in ORIGINAL scale:
       AB    OBP    SLG
0  24.650  0.375  0.455
1   2.418  0.036  0.018

Cluster label counts:
0    180
1    225
```

Figure 23: Cluster Centers

Cluster 0 has a centroid around 24.7 AB, .375 OBP and a .455 SLG Percentages. This describes hitters who got a moderate amount of postseason playing time and, on average, hit

decently well. They reach base more than a third of the time and show decent power. There are over 180 players in the group, nearly half of the entire postseason players. This reveals how the Dodgers have a smaller but more productive group that carries most of the offensive load.

Cluster 1 has a centroid around 2.4 AB, .036 OBP and .018 SLG, which is basically no playing time and almost no production. These are hitters who had only a couple of at-bats in the postseason and, on average, did not reach base or had enough hit power at all. There are over 225 players in the group which tells that a large share of postseason plate appearances come from brief, low-impact stints (pinch hitters, late-inning subs, call-ups).

Conclusions have been made that for the Dodgers, they use the same core group of post-season players in the play-offs. They do not make many changes to their group beyond subbing out players for specific situations.

K = 3

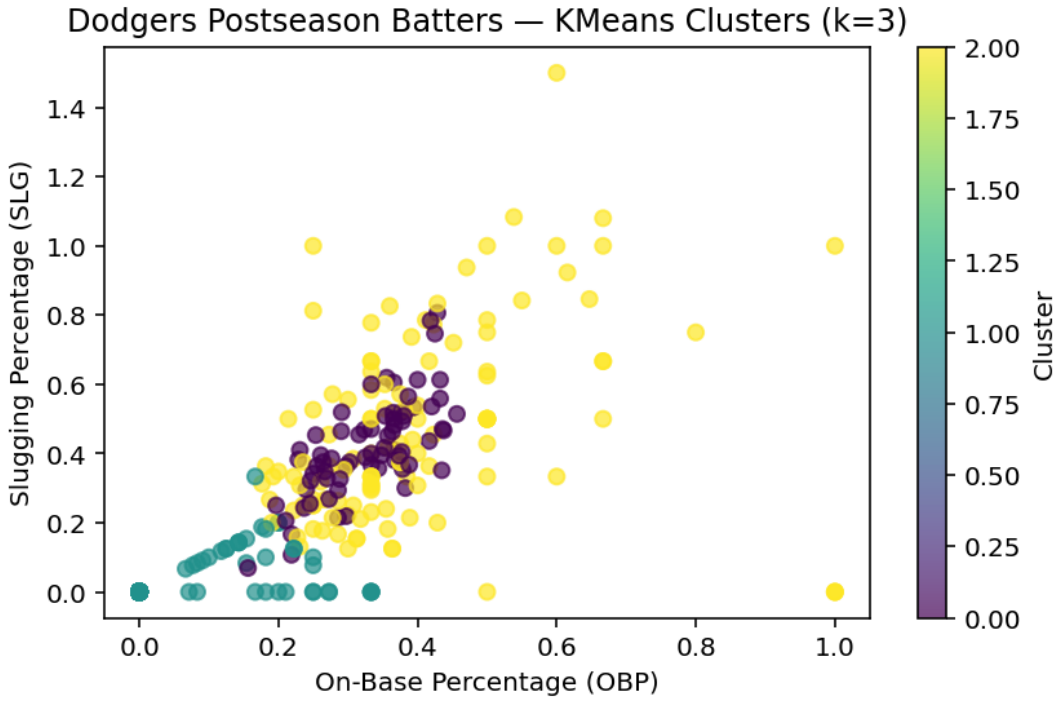


Figure 24: Cluster 0 vs Cluster 1 vs Cluster 2

```
Cluster centers (min-max SCALED AB, OBP, SLG):
      AB    OBP    SLG
0  0.624  0.329  0.279
1  0.032  0.031  0.009
2  0.168  0.398  0.310

Cluster centers in ORIGINAL scale:
       AB    OBP    SLG
0  44.271  0.329  0.419
1   2.305  0.031  0.014
2  11.957  0.398  0.466

Cluster label counts:
0     70
1    220
2    115
```

Figure 25: Cluster Centers

The k = 3 results reveal three types of postseason hitters now.

Cluster 0 has a centroid of 44.27 Abs, 0.329 OBP and .419 SLG. There are only 70 players on this list which shows that these are the hitters that the Dodgers have to rely on to gain

postseason success. Having so few players in comparison to the other clusters also reveals how hard it is to obtain postseason players that can be relied on a consistent basis. When the Dodgers have these types of players, their chances of success rise dramatically.

Cluster 1 has a centroid of 2.3 AB, 0.031 OBP and a .014 SLG. These make up the majority of players with over 220 players. These are basically the no-playing time/ no production group of the Dodgers. These are guys who only get a couple of AB's and, on average, don't reach base nor hit for much power. This cluster is more defined in comparison to k = 2. We get to see further proof that these are the type of players that have very few opportunities to raise their own stats due to their status in the game – bench players, pinch hitters, late-inning substitutions.

Cluster 2 has a centroid of 12 AB, .398 OBP and .466 SLG. As the second largest group with over 115 players, these players represent the part-time contributors. They don't have as much AB's but when they do play, they hit at an amazing rate. While they do have more OBP and SLG percentages than Cluster 0 (the heavy hitters of the Dodgers), it's explained by their lack of AB's. They have less AB's which inflates their stats in comparison to their counterparts. This just means that they can be relied on as role players.

K = 4

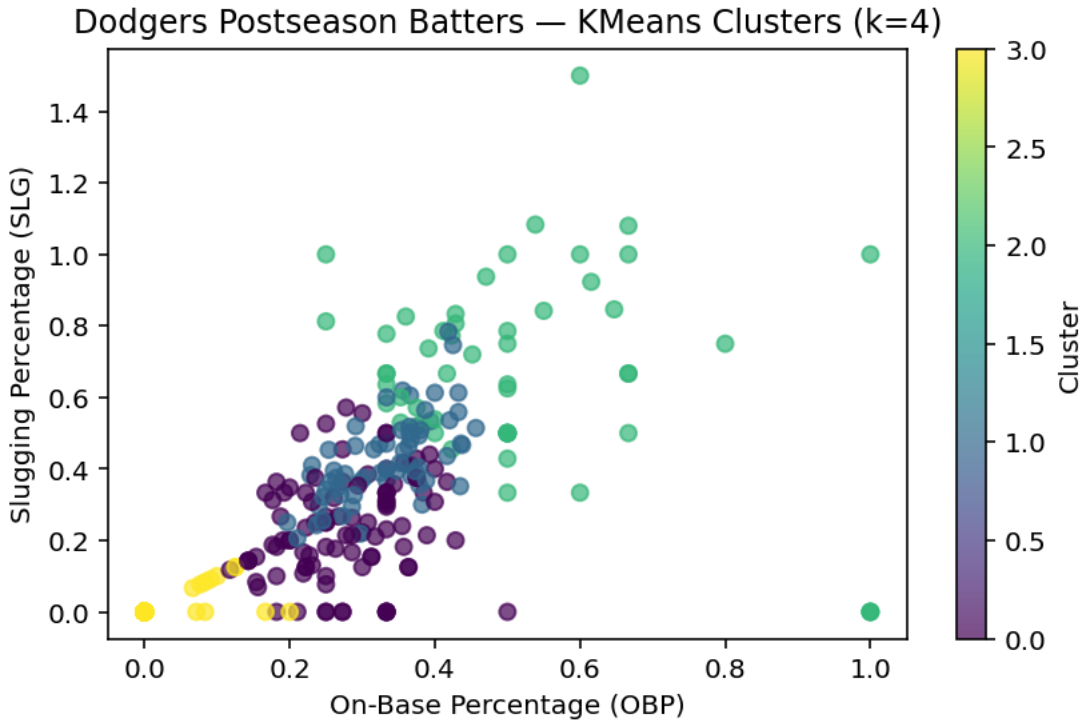


Figure 26: Cluster 0 vs 1 vs 2 vs 3

```
Cluster centers (min-max SCALED AB, OBP, SLG):
      AB    OBP    SLG
0  0.178  0.276  0.158
1  0.644  0.334  0.288
2  0.152  0.518  0.451
3  0.021  0.006  0.002

Cluster centers in ORIGINAL scale:
       AB    OBP    SLG
0  12.642  0.276  0.237
1  45.703  0.334  0.431
2  10.788  0.518  0.677
3   1.515  0.006  0.003

Cluster label counts:
0     95
1     64
2     52
3    194
```

Figure 27: Cluster Centers

With cluster of 4, we can differentiate even more – revealing more archetypes of post-season players.

Cluster 0 has an AB of 12 AB's, 0.276 OBP and a 0.237 SLG. With over 95 players, these hitters get a small to moderate number of AB's, but their production is at a below average both in getting on base and slugging power. Based on these stats, we can conclude that they are most likely bottom of the order bats or role players who were in the lineup but didn't produce much. On the scatter graph, they sit in the lower to middle region.

Cluster 1 has an AB of 45 AB's, 0.334 OBP and a 0.431 SLG. With only 64 players, these hitters are the high-volume regular hitters. They have the most AB's with a respectable average of OBP and SLG's. We can conclude that these players are the players who are near the top to the middle of the line-up – usually the players who help determine play-off success.

Cluster 2 has an AB of 10.8, 0.518 OBP and a 0.677 SLG. With only 52 players, these are guys who performed exceptionally well – strong on base percentages with a very high slugging percentage. These can be explained as players who are either role players who started extremely strong or as star Dodger players who only had a short postseason run. Limited runs can inflate stats. They appear near the top of the scatter plot.

Cluster 3 has an AB of 1.5 AB, .006 OBP and a 0.003 SLG. With the highest rate of players, a whopping 194, we can determine that these are the hitters that basically never got it going during the postseason. These are the most low-impact, short-stint players that can be explained as being pinch hitters, defensive subs or late-inning substitutions. They are near the very bottom of the scatter plot.

Conclusions

Overall, we can draw conclusions on the splits from k = 2 to k = 4. K = 2 had a simple split between contributors vs. non-contributors of the game of baseball. It was a defined separation that needed to be defined further for more analysis. Luckily, k = 3 gave us the opportunity to further define the contributors of the game. K = 3 divided up the groups into 3 distinct groups – regulars, efficient part-timers and non-contributors. Regulars are the high-volume hitters, efficient part-timers are the lower volume but very efficient hitters and the non-contributors are, basically, the non-contributors.

When dividing up the groups into 4, we start to split the part-timers into weak vs strong part-timers. We have the regular high-volume hitters, the strong part-timers, the weak part-timers and the non-contributors. The differences are smaller than they were before, but they help explain more archetypes of baseball that can help contribute as to why certain teams do better than others.

The Dodgers have a consistent group of guys that can contribute in the play-offs which explains why they have so much success in their runs. Having a large group of guys who don't contribute (the non-contributors) is completely normal in the game of baseball. Baseball is an extremely hard sport, and it's not expected to have over a 50% hitting rate. It's a game of inches not miles and having players that can step over that imaginary line of improvement can mean the difference between making a long post-season run and a short stint. Every player must contribute to a team's success.

Supervised Learning with Decision Trees

The first step to take when it comes to using Decision Tree Modeling is choosing a data source. The data that I will be using will be a continuation of the Dodger's postseason batting averages from 1990-2024.

A	B	C	D	E	F	G	H	I
season	playerNam	primaryPo	Age	AB	AVG	OBP	SLG	
1995	Antonio Os	P	0.090909	0	0	0	0	
1995	Billy Ashley	PH	0.2272727	0	0	1	0	
1995	Brett Butle	CF	0.8181818	0.2112676	0.2666666	0.2666666	0.066667	
1995	Chad Fonv	SS	0.1818181	0.1690140	0.5	0.5	0.125	
1995	Chris Gwyr	PH	0.4545454	0.0140845	0	0	0	
1995	Dave Hans	PH	0.2727272	0.042254	0.6666666	0.6666666	0.16666666666666666	
1995	Delino DeS	2B	0.2727272	0.1690140	0.25	0.3076923	0.0625	
1995	Eric Karros	1B	0.3181818	0.1690140	0.5	0.5384615	0.2708333333333333	
1995	Hideo Nom	P	0.3181818	0.0281690	0	0	0	
1995	Ismael Val	P	0.090909	0.042254	0	0	0	
1995	John Cumr	P	0.2727272	0	0	0	0	
1995	Jose Offerr	PR	0.2727272	0	0	0	0	
1995	Kevin Tapa	P	0.5	0	0	0	0	
1995	Mark Guth	P	0.4545454	0	0	0	0	
1995	Mike Piazz	C	0.3181818	0.1971830	0.2142857	0.2142857	0.125	
1995	Mitch Web	PH	0.7272727	0.0281690	0	0	0	
1995	Pedro Asta	P	0.2727272	0	0	0	0	
1995	Ramon Ma	P	0.3181818	0.0140845	0	0	0	
1995	Raul Mond	RF	0.1818181	0.1267605	0.2222222	0.3	0.055556	
1995	Roberto Ke	LF	0.5	0.1549295	0.3636363	0.4166666	0.090909	
1995	Tim Wallac	3B	0.8181818	0.1690140	0.083333	0.1538461	0.020833333333333325	
1995	Todd Holla	RF	0.090909	0.0281690	0	0	0	
1996	Antonio Os	P	0.1363636	0	0	0	0	

Figure 28: Dodger's Post-Season Batting Averages

In order for the data to be used for Sklearn Decision Tree Classification, the dataset must be labeled and only have quantitative columns for the features. The next step is to ensure that any data that is qualitative must be removed. With my data, the playerName and primaryPosition are removed from the dataset.

```
   season       Age        AB       AVG       OBP       SLG
0    1995  0.090909  0.000000  0.000000  0.000000  0.000000
1    1995  0.227273  0.000000  0.000000  1.000000  0.000000
2    1995  0.818182  0.211268  0.266667  0.266667  0.066667
3    1995  0.181818  0.169014  0.500000  0.500000  0.125000
4    1995  0.454545  0.014085  0.000000  0.000000  0.000000
```

Figure 29: Removal of Qualitative Data

The next step is to choose a label column that my data will predict. The one that I will predict will be defining what a good batter consists of. To do so, I will create 2 labels – 0 (worse batters), 1 (better batters). They will be defined by their slugging averages. The label will be called Good Hitters. After adjusting to these new rows, we get an equal amount of 0 and 1.

```
Label value counts (0=bad, 1=good):
GoodHitter
0    203
1    203
```

Figure 30: Value Count of 0 and 1

This shows that our dataset is perfectly balanced.

```
Shapes:
TrainingData: (284, 6)
TrainingLabel: (284,)
TestingData: (122, 6)
TestingLabel: (122,)

Train accuracy: 1.0
Test  accuracy: 1.0
```

Figure 31: TrainingData, TrainingLabel, TestingData and TestingLabel

The Training Data: (284, 6) shows that there are 284 players in the training set. Each player is described by 6 quantitative variables – season, Age, At Bats, Averages, On-Base- Percentage and Slugging Percentage. The Training Label shows that there are 284 labels for each player. The

Testing Data (122, 6) reveals that there are 122 players from the testing set with the same 6 features and 122 labels.

The next step is to create visualizations of our training data to offer information about the data.

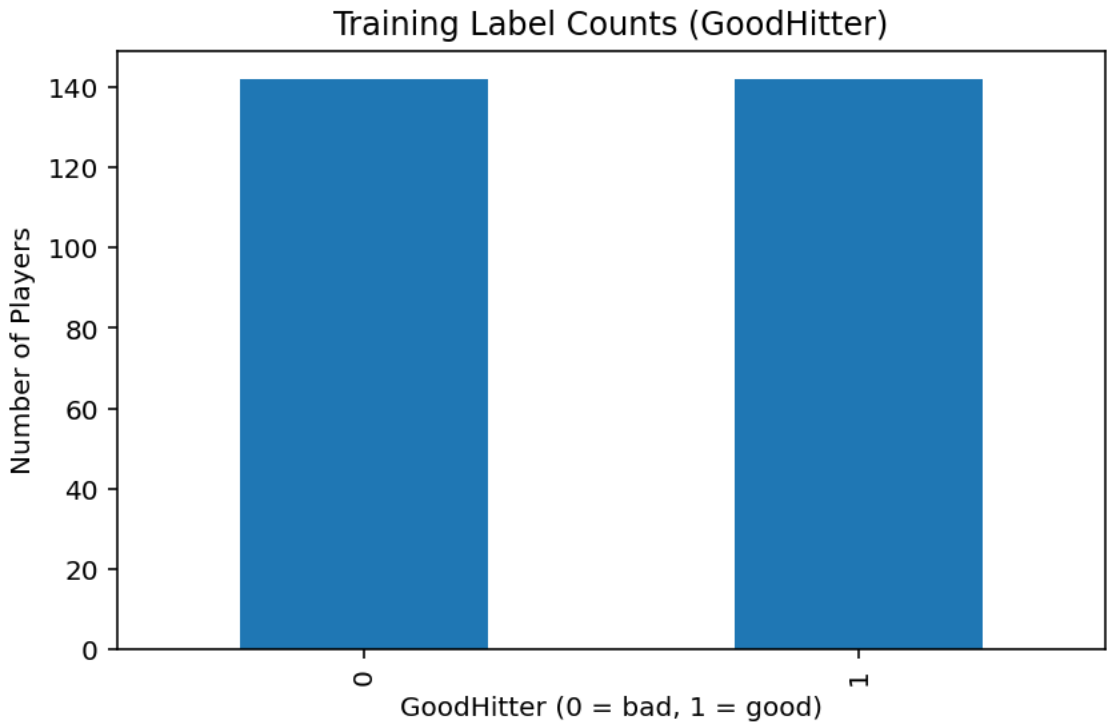


Figure 32: Training Label

The bar chart reveals that the training data is perfectly balanced between bad and good hitters, with almost the same amount of players in each group. Because the GoodHitters's are based on a median split of slugging percentage, half of the batters are classified as below-median and half at-or above median in terms of power production. A balanced label is helpful because the Decision Tree will not be trained on a dataset that favors one class, therefore, excluding any type of bias.

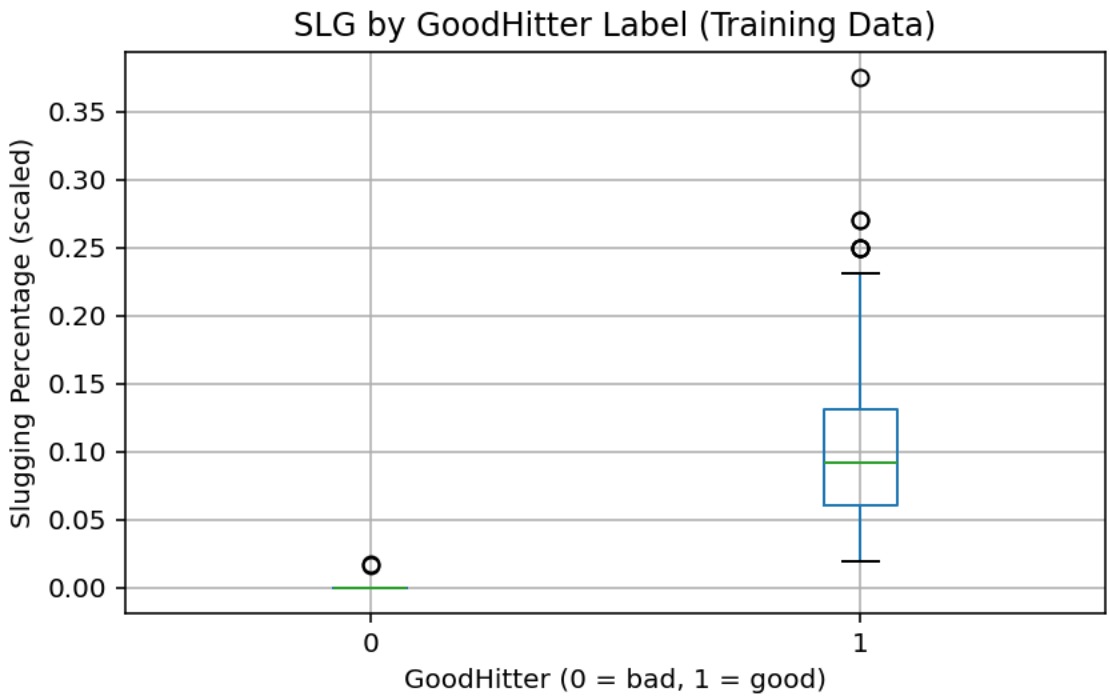


Figure 33: Boxplot SLG

This plot shows that the Goodhitter label separates players be their slugging percentage. Batters labeled 0 have SLG values that are essentially at the very bottom of the scaled range while batters labeled 1 have much higher SLG values with a median well above the 0 group and a wider spread into the higher end of the scale. This strong separation can help explain why the Decision Tree can predict the labels accurately.

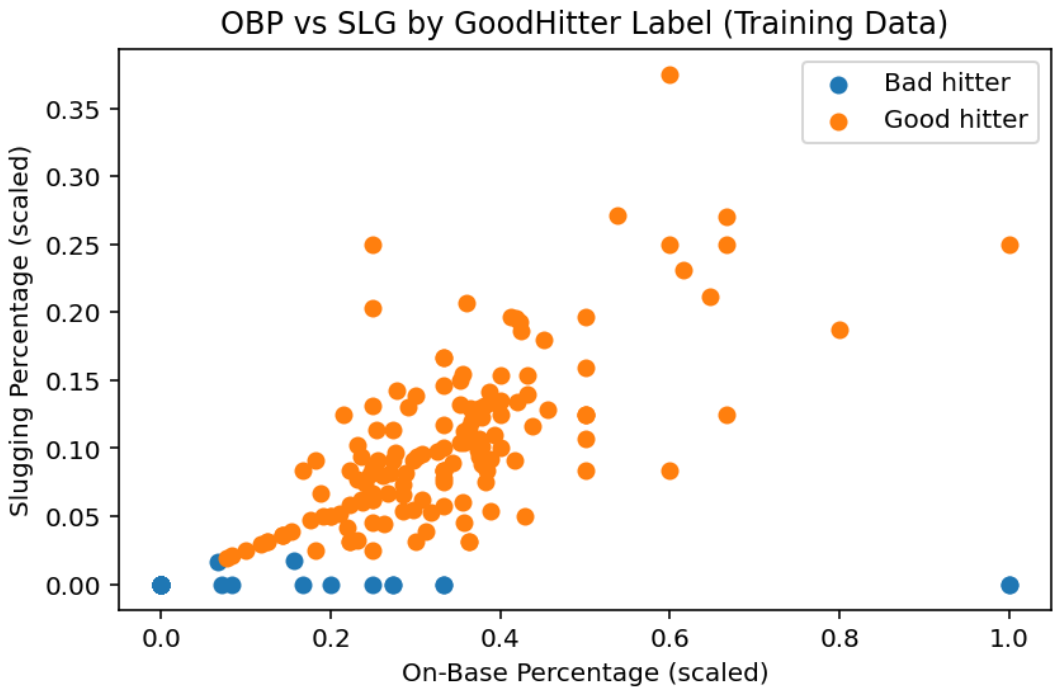


Figure 34: Scatter Plot

This scatter plot suggests that on-base ability and power are strongly linked for the players labeled as good hitters, while bad hitters tend to cluster in a low-performance region on both metrics. Most of the good hitters are well above the median slugging percentage which shows what exactly is needed to be defined as a good hitter.

After training the Decision Tree on the Training Data and then testing it on the Testing Data, I evaluated the performance using a confusion matrix.

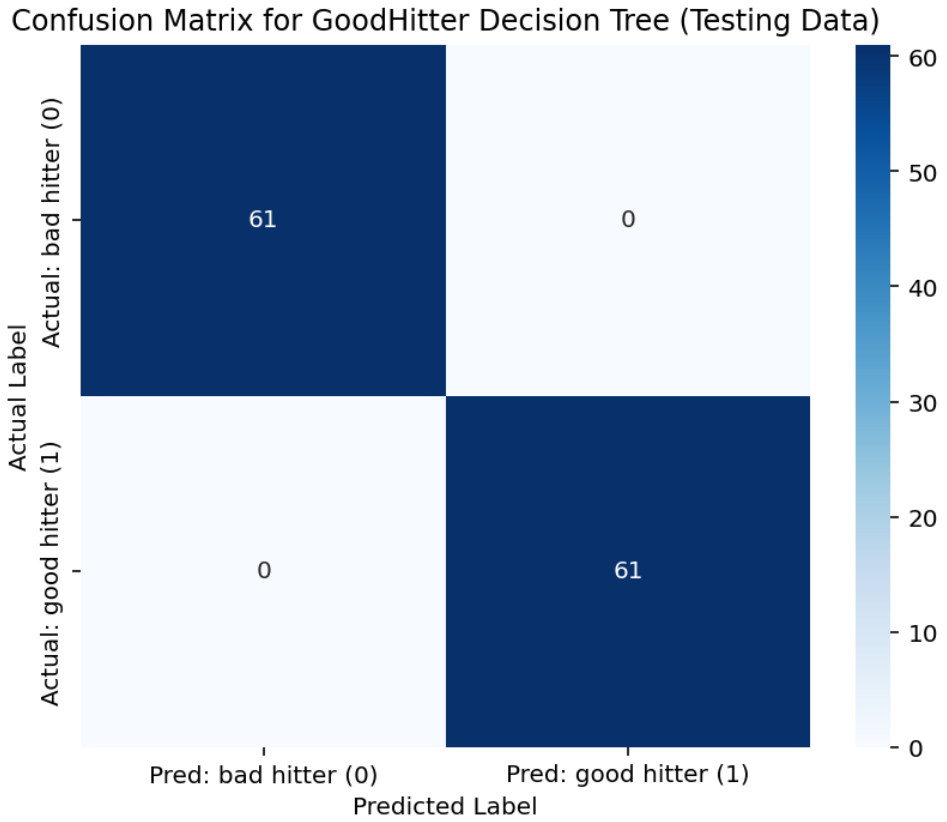


Figure 35: Confusion Matrix

The matrix shows that every batter in the test set was placed on the diagonal cells (bad hitters predicted as 0 and good hitters predicted as 1), with 0 counts in the off-diagonal cells. This means that the model made no misclassifications on the Testing Data, yielding an overall accuracy of 100%. This is expected because the GoodHitter label was defined directly from the slugging percentage (SLG) and the SLG was one of the predictor variables. The Decision Tree basically relearned the same threshold rule used to create the label.

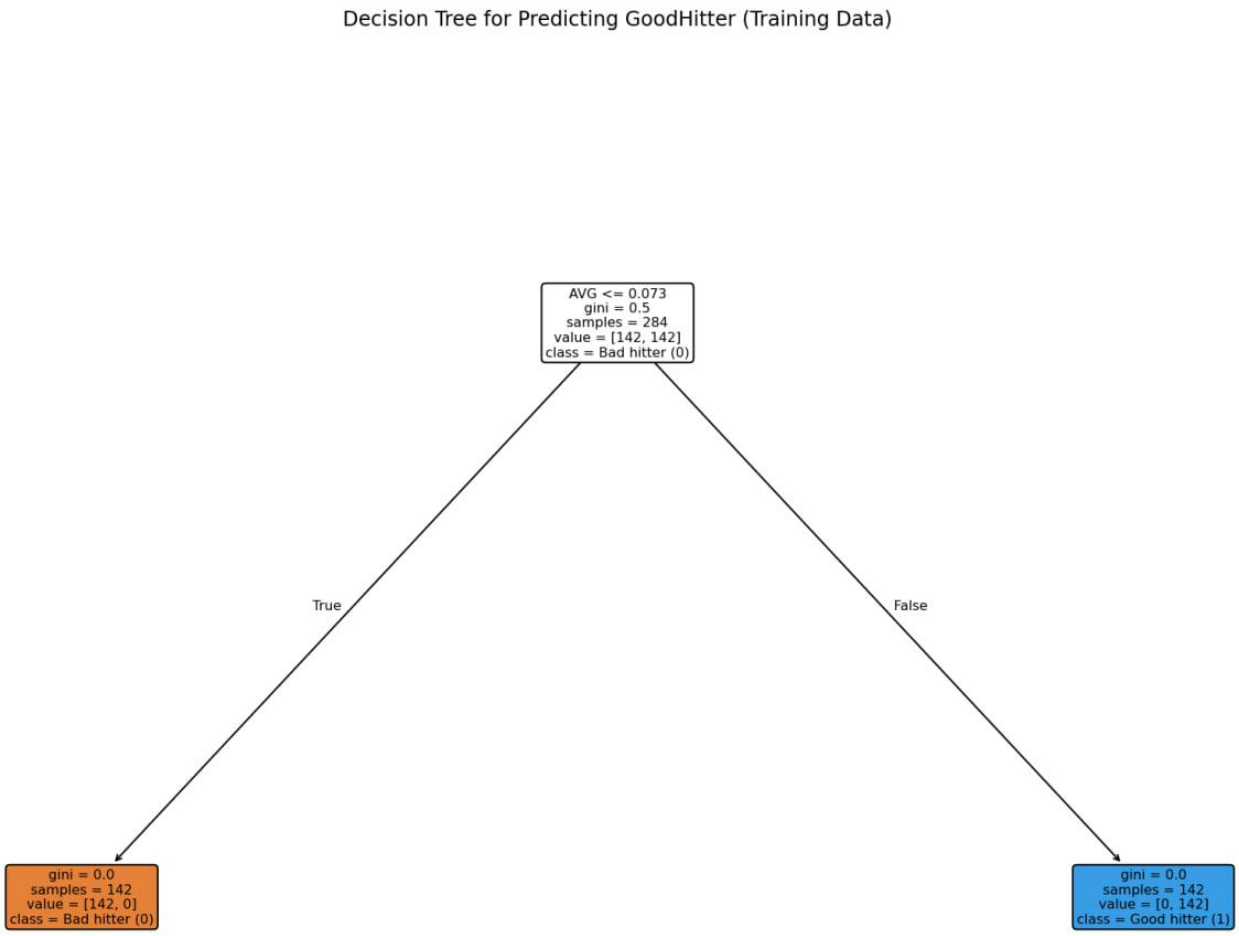


Figure 36: Decision Tree

This Decision Tree learned a simple rule: it splits hitters only based on batting average (AVG). If a player's AVG is at or below 0.073, the model predicts they are a bad hitter; if it is a higher AVF, the model will create a good hitter. Each leaf node is accurate. All bad on the left side and all good on the right side.

Conclusion

For this project, I used a decision tree model to separate postseason Dodgers hitters into two groups: "good hitters" and "bad hitters." I defined good hitters as players whose slugging percentage (a power-hitting stat that rewards extra-base hits) was at or above the middle value in the data, and bad hitters as those below that point. This gave me a very balanced dataset – about

half of the players in each group – and I trained the model on part of the data and tested it on the rest to see how well it could tell the difference.

The model ended up learning a very simple rule. It basically says: “If a player’s batting average is above a small cutoff, call them a good hitter; if it’s below that cutoff, call them a bad hitter.” Using just this one split, the tree correctly classified every player in the test set, so it agreed 100% with the way I had labeled the hitters. The visualizations support this: the “good hitter” group has clearly higher slugging percentages and generally higher on-base percentages, while the “bad hitter” group is clustered at the low end for both stats.

Overall, the analysis tells us that in this postseason dataset, basic hitting results move together: players who get more hits also tend to hit for more power and reach base more often. Because of that, the model doesn’t need a really complicated set of rules to separate strong and weak bats- it can draw a straight line to define weak and strong batters. This kind of result is useful because it confirms what I, have personally, defined as a good hitter to what we know for offensive productions.

Conclusions



Figure 37: Dodgers Win

Over the last three and a half decade, the Dodgers have relied on very different types of hitters to carry their offense in October. By looking at how often players came to the plate, how often they reached base, how much power they showed, and how old they were at the time, we were able to see clear patterns in the way postseason lineups have actually been built. Instead of just thinking about “good” or “bad” hitters, the data suggested distinct roles: everyday stars who do the heavy lifting, surprise bats who make a big impact in limited chances, steady role players, and depth pieces who barely get to swing.

One of the clearest lessons is how important the “everyday anchors” are. These are the hitters who rack up a lot of at-bats in a single postseason run and still manage to post solid numbers in getting on base and hitting for extra bases. When these players are productive, the

offense looks deeper and more stable, because they are in the middle of almost every rally. The flipside is that when this group struggles, it's very hard for the rest of the lineup to make up the difference, no matter how many interesting supporting pieces the team has around them.

A second important group is what we might call the "impact bats." These are hitters who don't always get the same number of chances as the everyday starters, but when they do play, their numbers are extremely strong. They often show up as younger players or part-time starters who bring a lot of power or on-base skill in a smaller number of opportunities. For a front office or coaching staff, this suggests that finding and trusting these types of hitters can be a real advantage in October. Giving them just a few extra plate appearances at the right time could swing a game or even a series, especially when the usual stars are being pitched around or are having a cold stretch.

Finally, the data also highlighted how many players fall into more limited roles. There are "role players" who get a modest number of chances and produce average-looking results, and there is also a long list of players who barely get on the field at all. For those players, their numbers are often so small that it's hard to judge their true ability from postseason stats alone. This is an important reminder not to overreact to tiny samples. A player who went hitless in just a couple of at-bats in October should not automatically be labeled as a poor performer; the data mainly tells us that their role was small, not that they are incapable of contributing.

Overall, the story that emerges is that the Dodgers' postseason offenses are built around a core of everyday hitters, boosted by a smaller group of high-impact bats, and surrounded by role players and depth pieces whose opportunities and results are much more limited. For decision-makers, this means that October success depends not just on having "big names," but on making sure those key everyday hitters are supported by at least a few impact bats who can change a

game in fewer chances. It also means being careful about how postseason numbers are interpreted, especially for bench players with very few at-bats. Used wisely, these patterns can help the team design lineups, manage playing time, and think more clearly about which types of hitters they need to prioritize going into future playoff runs.

Reference

Bloomberg. (2024). *The "GOAT" of basketball dunking"* [Photograph]. Bloomberg. https://www.bloomberg.com/en/news/thp/2024-04-24/shohei-ohtani-hits-450-foot-homer-into-second-deck-at-nationals-park-in-dodgers-4-1-win

josephvm. (n.d.). *MLB Game Data* [Data set]. Kaggle. https://www.kaggle.com/datasets/josephvm/mlb-game-data

Los Angeles Dodgers Batting & Pitching (1958–2023) — Kaggle: https://www.kaggle.com/datasets/mattop/los-angeles-dodgers-batting-and-pitching-1958-2023

Major League Baseball. (2024). *MLB StatsAPI: Postseason schedule and box scores (2024 season)* [Data set]. MLB Advanced Media. https://statsapi.mlb.com/api/v1